COMP3710 / COMP7710 PATTERN RECOGNITION

HipMRI prostate segmentation

Does a residual 3D U-Net improve apex and base contours over a slice-wise 2D U-Net?

Project: Medical image analysis **Difficulty:** Hard - 3D Improved U-Net

Evidence reviewed: repository README, code, saved run records, paired case metrics, and five error panels

Abstract

This project evaluates whether a small residual 3D U-Net with deep supervision gives more useful prostate contours than a slice-wise 2D U-Net on longitudinal T2-weighted pelvic MRI. The local HipMRI release contains 211 paired image and mask volumes from 38 patients. A patient-disjoint split reserved 23 visits from five patients for testing. In the archived full-field experiment, patient-balanced native prostate Dice was 0.694 for 3D versus 0.549 for 2D; the descriptive paired difference was +0.144 (95% bootstrap interval +0.020 to +0.283). The 3D model improved mean base Dice and HD95 in this controlled configuration, but used about three times as many parameters and 24 ms more model-only CPU time. A stronger cropped 2D comparison described in the README reports Dice 0.698, but its underlying run files are absent from the supplied folder. The early pilot exposed the test split; therefore all test outcomes are exploratory, and no clinical or causal advantage is established.

Decision: Retain both models as research drafts requiring specialist review. The archived full-field comparison favors 3D, while the README-reported tuned 2D result prevents a robust claim that 3D is the better deployment choice.

Evidence labels used in this report

Archived: values directly checked against JSON/CSV in this folder. **README-reported:** values stated in project prose or figures without the associated run output. These are identified where used.

1. Feasibility review

User need. A radiotherapy contouring specialist could use a draft prostate mask to speed review, provided that missed apex/base tissue and overlap with bladder or rectum remain visible. The intended output is an editable research contour, not autonomous treatment planning. The central question is whether through-plane image information improves clinically relevant edge-slice quality enough to justify higher compute cost [1-3].

Inputs and outputs. The system takes one 3D T2-weighted NIfTI MRI and an aligned expert semantic mask for training or evaluation. It predicts a binary prostate mask. Native semantic labels are retained to quantify false-positive voxels in bladder and rectum. The implementation also produces patient-level metrics and visual error panels.

Acceptance criteria and observed status

Criterion	Observed status
Patient-disjoint evaluation	Met: 27/6/5 patients in train/validation/test.
Downsampled 3D Dice at least 0.70	Met in archived test run: 0.757 patient mean on network grid.
Native contour quality	3D native Dice 0.694; apex 0.348 and base 0.636.
Latency below 5 s; GPU memory below 16 GiB	CPU model call 74 ms; target-GPU memory and latency unmeasured.
Failure review	Five distinct-patient panels with error overlays and case notes.

Risks and fallback. The prostate occupies roughly 0.10% of native voxels, so class imbalance and loss of detail during resampling are central risks. A larger 3D grid may retain boundaries but requires a new validation-only resource study. The README describes a validation-selected central-crop 2D model as a lower-cost fallback; its test metrics need the missing run files before they can be treated as independently verified.

Budget and decision gate. The measured CPU experiment uses a 32 x 64 x 64 grid and four starting channels. Before any operational recommendation, rerun on course hardware, record peak allocated/reserved GPU memory and per-case latency, evaluate an independent patient cohort, obtain specialist adjudication, and assess native-space dosimetric consequences.

2. Data and methods

2.1 Data and split

The downloaded CSIRO HipMRI release [4] contains 211 aligned image/mask visits from 38 patients. The saved audit reports 210 volumes of shape 256 x 256 x 128 and one of 256 x 256 x 144; prostate label 5 occurs in every mask. Course-tutor confirmation of label IDs 5 (prostate), 3 (bladder), and 4 (rectum) is recorded in the README. The fixed-seed manifest split groups all visits from each patient: 154 training visits from 27 patients, 34 validation visits from six, and 23 test visits from five. The test patients are disjoint from development patients.

2.2 Preprocessing

NIfTI image and mask geometry is audited before training. Volumes are reoriented to canonical axes. Nonzero MRI intensities are clipped to the 0.5th and 99.5th percentiles and standardized within each case. MRI is resized with trilinear interpolation; integer labels use nearest-neighbor interpolation. The full-field experiment uses a 32 x 64 x 64 depth-height-width grid. Binary prostate targets are derived from semantic label 5, while the original mask supports bladder and rectum leakage analysis. The reported runs used no augmentation.

2.3 Models and optimization

The baseline is a 2D U-Net applied independently to axial slices [1]. The proposed model uses 3D convolutions, residual blocks, skip connections, and two auxiliary decoder heads for deep supervision [2, 3]. With base width four, the 3D encoder uses 4, 8, 16, and 32 channels and three pooling stages. Only the final full-grid output is used at inference. Both models use weighted cross-entropy plus soft prostate Dice loss; the auxiliary 3D losses have weights 0.25 and 0.125. AdamW uses learning rate 0.001, foreground weight 20, and up to ten epochs in the archived selected run. The 2D trainer oversamples foreground slices; the 3D trainer samples volumes.

> **Interpretation constraint:** The models differ in dimensionality, residual blocks, deep supervision, and training sample selection. A score difference cannot be attributed to 3D context alone.

2.4 Selection and locked evaluation

Checkpoints are chosen by patient-balanced validation Dice. The archived full-field weight-20 run selected its 2D checkpoint at epoch 6 and 3D checkpoint at epoch 5. Final predictions were restored to the canonical native mask grid and scored on the same test cases. A one-time evaluation gate records the attempt and refuses a second test read for that run. However, a preceding one-epoch software pilot had already exposed this test split; the gate improves later workflow discipline but does not make the split blind.

Pipeline at a glance

1. Input	2. Split	3. Fit	4. Evaluate
Aligned MRI + semantic mask	Patient-disjoint 27 / 6 / 5	2D U-Net and residual 3D U-Net	Native Dice, boundaries, leakage, latency

3. Evaluation design

3.1 Metrics

Overlap: prostate Dice and intersection-over-union (IoU) are measured both on the network grid and after nearest-neighbor restoration to native space. Background Dice is recorded but is not a useful primary measure because background dominates the volume. **Boundary:** apex and base Dice use the first and last 20% of expert-annotated gland slices; HD95 measures the 95th-percentile surface distance in millimeters. **Anatomical leakage:** the fraction of bladder or rectum voxels predicted as prostate and false positives beyond the gland slice extent. **Resources:** trainable parameters, model-only CPU inference time after preprocessing, and fit time. Target-GPU memory is not available.

3.2 Unit of analysis

The 23 test visits belong to only five patients. Headline values average visits within each patient and then average across patients, so patients with more weekly visits do not dominate. The paired 95% intervals resample patients 10,000 times and are descriptive rather than a guarantee of generalization.

3.3 Available validation evidence

Archived run	2D validation Dice	3D validation Dice	Status
Full field, weight 1	0.606	0.597	Validation only
Full field, weight 20	0.631	0.684	Selected full-field run

The README describes two additional central-crop validation candidates, including a 2D model with validation Dice 0.708. Their machine-readable run records are not present in the supplied folder, so the table above includes only archived runs. The full-field weight-20 result was selected for the verified 3D comparison.

3.4 Image review protocol

Five distinct patients were chosen for full-field error panels. Each shows apex, mid-gland, and base MRI slices; expert contour; 2D and 3D contours; and false-positive (red) / false-negative (blue) overlays. The panels show the resampled network grid, while the numeric case Dice in Appendix A is calculated on native masks. Visual observations are hypotheses about failure mode until a contouring specialist adjudicates them.

4. Archived full-field results

The table below comes from the selected run's saved *results.json* and *paired_summary.json*. Both models use the same patient split and full-field preprocessing. Values are patient-balanced unless otherwise stated.

Metric	2D U-Net	3D residual U-Net
Native prostate Dice	0.549	0.694
Native prostate IoU	0.390	0.542
Network-grid prostate Dice	0.610	0.757
Native apex Dice	0.301	0.348
Native base Dice	0.449	0.636
Native HD95 (mm; lower is better)	14.41	8.17
False-positive fraction in bladder	2.28%	0.97%
False-positive fraction in rectum	0.341%	0.261%
Parameters	30,782	93,358
Model-only CPU latency	50.1 ms	74.0 ms
Peak target-GPU memory	Unmeasured	Unmeasured

The paired native Dice difference (3D minus 2D) is **+0.144**, with a descriptive five-patient bootstrap interval of **+0.020 to +0.283**. Base Dice rises by +0.188 (interval -0.018 to +0.508), while HD95 falls by 6.23 mm (interval -10.46 to -2.01 mm). The network-grid 3D Dice of 0.757 exceeds the draft brief's 0.70 target, but native Dice is 0.694. These are internal, previously exposed test data.

5. Patient variation and robustness

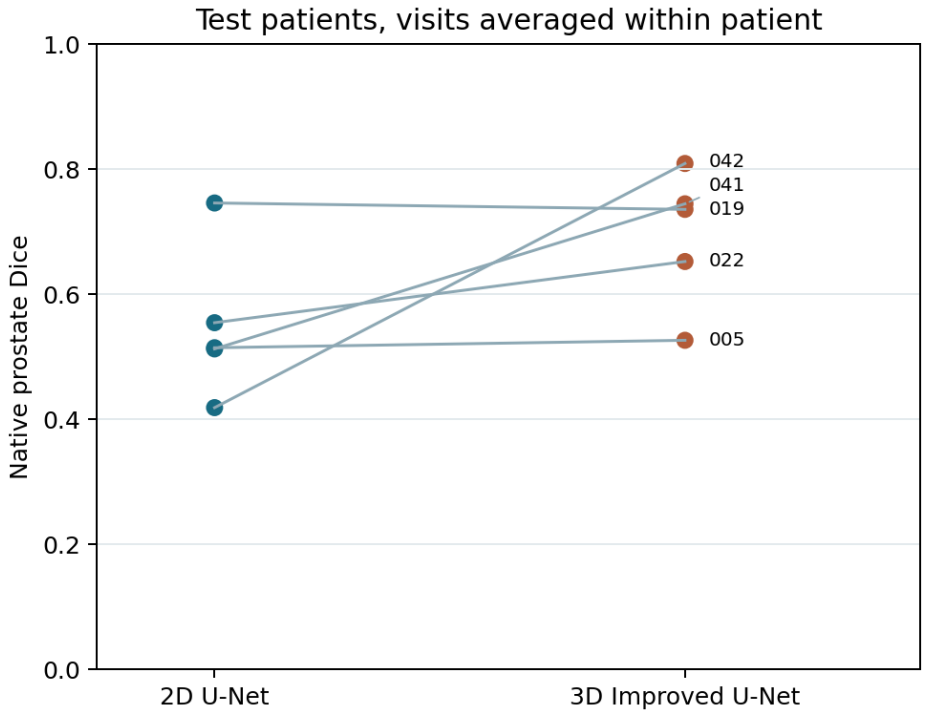


Figure 1. Full-field native-grid prostate Dice by test patient, with weekly visits averaged within patient. Lines connect each patient's 2D and 3D results. Source: project figure and archived per_case.csv.

The figure shows four patients improving under the full-field 3D model and one patient (019) worsening. This variation matters more than a single mean, because five patients provide limited evidence about future cases.

README-reported cropped 2D check

The README states that a central-crop 2D model, selected independently on validation data, achieved native patient-balanced Dice 0.698 against 0.694 for the full-field 3D model. It also reports apex Dice 0.388 versus 0.348, base Dice 0.645 versus 0.636, HD95 9.11 versus 8.17 mm, and model-only CPU latency 52.1 versus 74.0 ms. The claimed tuned-run JSON and CSV files are **absent from this supplied folder**. These values are therefore reported as project claims, not independently verified measurements in this audit.

> **Implication if the reported cropped 2D values are confirmed:** The selected 3D model has no native Dice or apex/base overlap advantage over the stronger 2D comparator. The archive must include the tuned 2D run outputs to make this comparison reproducible.

6. Discussion and recommendation

The archived common-preprocessing comparison supports a local improvement in native Dice and HD95 for the 3D model. It does not isolate the effect of volumetric context, because architecture, auxiliary supervision, and sampling also change. The README-reported stronger 2D result, if confirmed, removes the apparent native Dice benefit. CPU inference is still fast in this small experiment, but the 3D model uses 93,358 versus 30,782 parameters, and target-GPU memory and end-to-end workflow time remain unmeasured.

The test cohort contains five patients, and an earlier pilot exposed it before final selection. This prevents an untouched generalization estimate. The 32 x 64 x 64 grid can erase small apex/base structures. Figures reveal missed edge-slice tissue and organ overlap that whole-volume Dice would conceal. No specialist reviewed the predictions, and there is no dosimetric or independent-cohort assessment.

Engineering decision: retain both systems as research drafts. Require human review of every proposed contour, especially apex/base slices and bladder/rectum overlap. Treat the cheaper cropped 2D model as the next comparator once its outputs are archived. Before considering 3D adoption, benchmark both at higher resolution on the target GPU, record peak memory and per-case latency, evaluate unseen patients, and obtain specialist and dosimetric review.

7. Reproducibility and disclosure

The archived selected run records CPU PyTorch, seed 3710, a 32 x 64 x 64 grid, width four, learning rate 0.001, foreground weight 20, ten epochs, and no augmentation. The repository includes manifest construction, audit, training, one-time test evaluation, metrics, plotting, and synthetic tests. Raw images and model weights are excluded from Git. The full-field results are traceable to saved JSON/CSV files; the tuned crop result needs its missing run files added to the evidence archive.

AI usage. The project README states that OpenAI Codex assisted with design, debugging, and writing. Codex also drafted and formatted this PDF from the repository and checked archived numeric claims against the saved run files. The submitting student should review all methods, figures, citations, and statements of contribution before using this report.

References and internal evidence

[1] O. Ronneberger, P. Fischer, and T. Brox. U-Net: Convolutional Networks for Biomedical Image Segmentation. MICCAI, 2015. arXiv:1505.04597.

[2] O. Cicek et al. 3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation. MICCAI, 2016. arXiv:1606.06650.

[3] F. Isensee et al. Brain Tumor Segmentation and Radiomics Survival Prediction: Contribution to the BRATS 2017 Challenge. 2018. arXiv:1802.10508.

[4] CSIRO. Labelled weekly MR images of the male pelvis. Data collection, 2021. doi:10.25919/45t8-p065.

[5] COMP3710/COMP7710 Pattern Recognition Report, 2026 draft v2.2, supplied course brief.

[6] Project records: README.md; runs/hipmri_native/data_audit.json; runs/hipmri_native_weight20/{split.json,results.json,paired_summary.json,per_case.csv}; PILOT_NOTES.md.

8. Reproduction recipe

Use Python 3.12 from the project directory. The pinned requirements are PyTorch 2.11.0, NumPy 2.4.3, nibabel 5.4.2, SciPy 1.17.1, Matplotlib 3.10.8, and pytest 9.0.2. The course cluster needs a PyTorch build matching its CUDA driver. Raw NIfTI data and model weights remain outside Git.

Prepare and audit the native data

Set `HIPMRI_ROOT` to the downloaded CSIRO release. The exact source subdirectories are `semantic_MRs_anon` and `semantic_labels_anon`.

```
python -m pip install -r requirements.txt
python -m pytest -q tests
python prepare_manifest.py --image-dir "$HIPMRI_ROOT/semantic_MRs_anon" \
--mask-dir "$HIPMRI_ROOT/semantic_labels_anon" \
--image-suffix _LFOV --mask-suffix _SEMANTIC_LFOV \
--patient-regex "^Case_([0-9]+)_Week[0-9]+$" \
--out runs/hipmri_native/manifest.csv
python audit_data.py --manifest runs/hipmri_native/manifest.csv \
--out runs/hipmri_native/data_audit.json
```

Train the archived full-field configuration in a fresh output directory

```
python train.py --manifest runs/hipmri_native/manifest.csv \
--prostate-label 5 --bladder-label 3 --rectum-label 4 \
--out runs/new_fullfield --device cpu --seed 3710 \
--size 32 64 64 --width-2d 4 --width-3d 4 \
--lr 0.001 --foreground-weight 20 --epochs 10 --patience 10 \
--batch-2d 16 --batch-3d 1 --validation-only
```

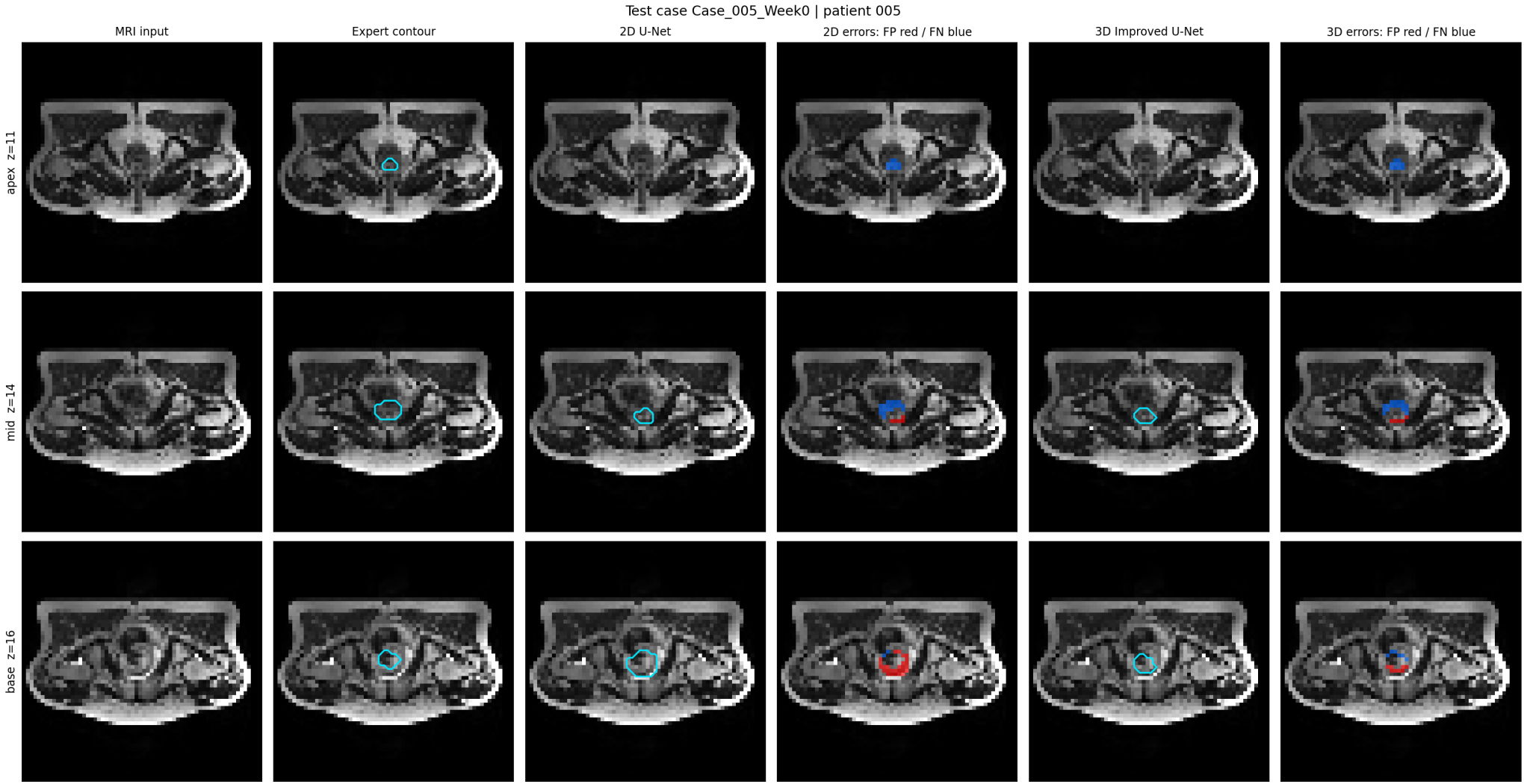
The README documents the separate one-time `evaluate_test.py` gate and plotting commands. Do not re-evaluate the saved run or use this already exposed test split for further tuning. For target-GPU profiling, use a new validation-only run with `--device cuda` and record device name, peak allocated and reserved memory, and per-case latency.

Code map

Files	Responsibility
modules.py, dataset.py	Networks; NIfTI loading, preprocessing, and patient split.
train.py	Validation-gated training, checkpoint selection, profiling.
evaluate_test.py, metrics.py	One-time test evaluation and native/grid metrics.
predict.py, analyze_results.py	Five-case overlays, paired summaries, and figures.

Appendix A.1 | Case_005_Week0

Native Dice: 2D to 3D = 0.259 to 0.369 | full-field models only

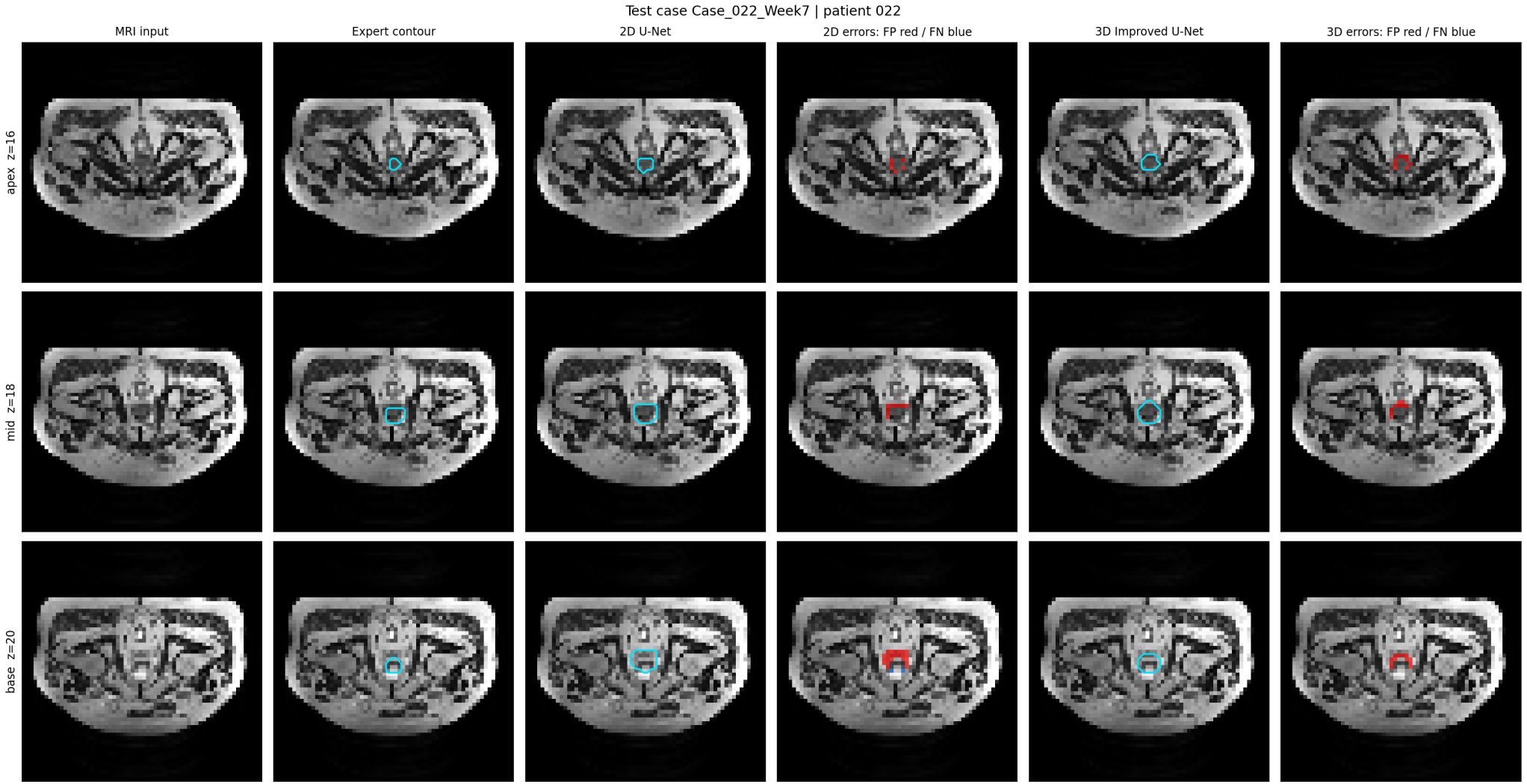


Both models miss the small apex; 3D apex Dice is 0. At mid-gland, 3D underfills the contour. Review any empty apex slice or broad blue missed region.

Panel source: repository figure. Red = false positive; blue = missed expert prostate. Image-grid panel; numeric Dice is native-grid.

Appendix A.2 | Case_022_Week7

Native Dice: 2D to 3D = 0.532 to 0.684 | full-field models only

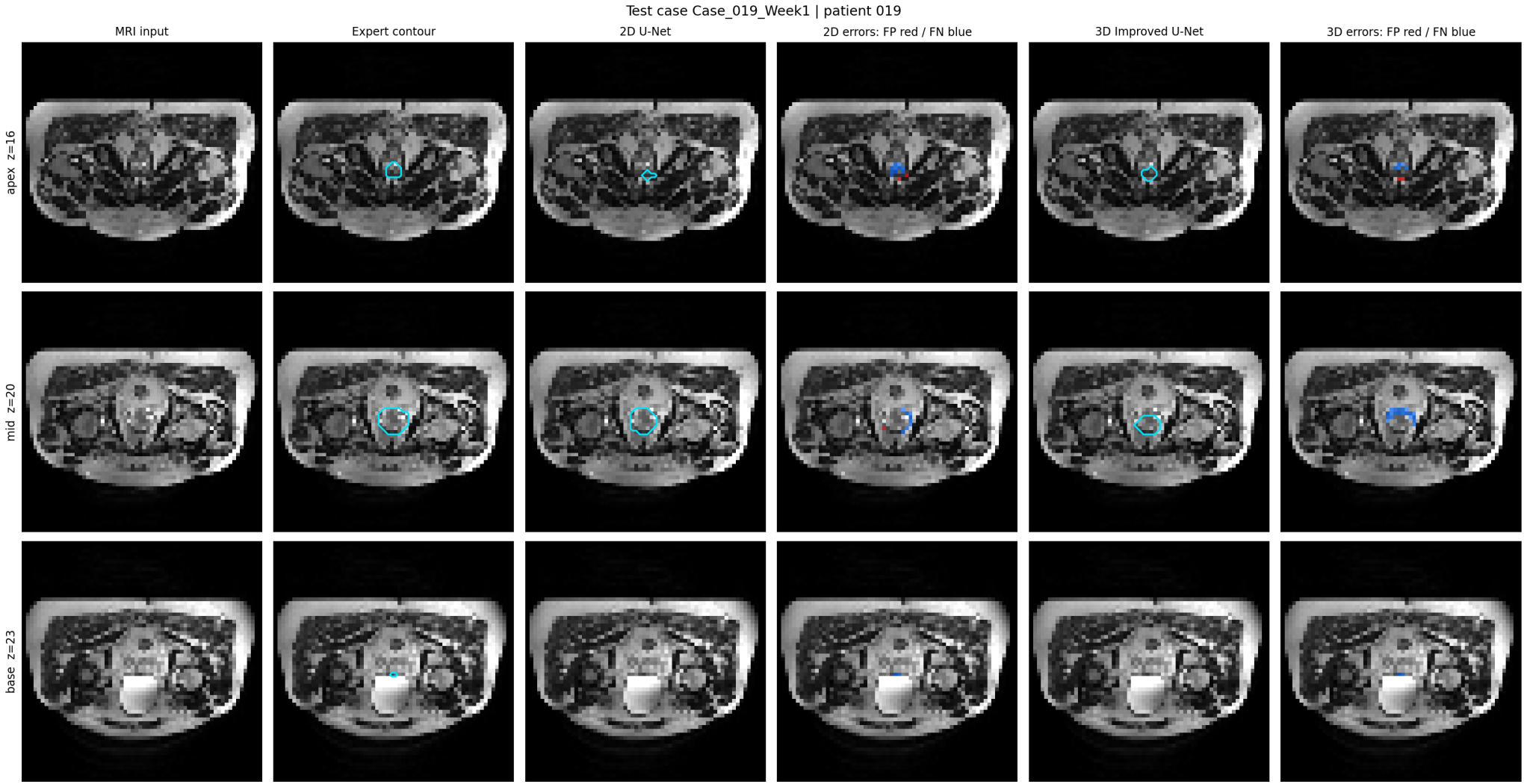


3D improves mid-gland overlap but still overlaps bladder. Its native bladder false-positive fraction is 4.6% for this visit. Review organ contact and base extension.

Panel source: repository figure. Red = false positive; blue = missed expert prostate. Image-grid panel; numeric Dice is native-grid.

Appendix A.3 | Case_019_Week1

Native Dice: 2D to 3D = 0.718 to 0.609 | full-field models only

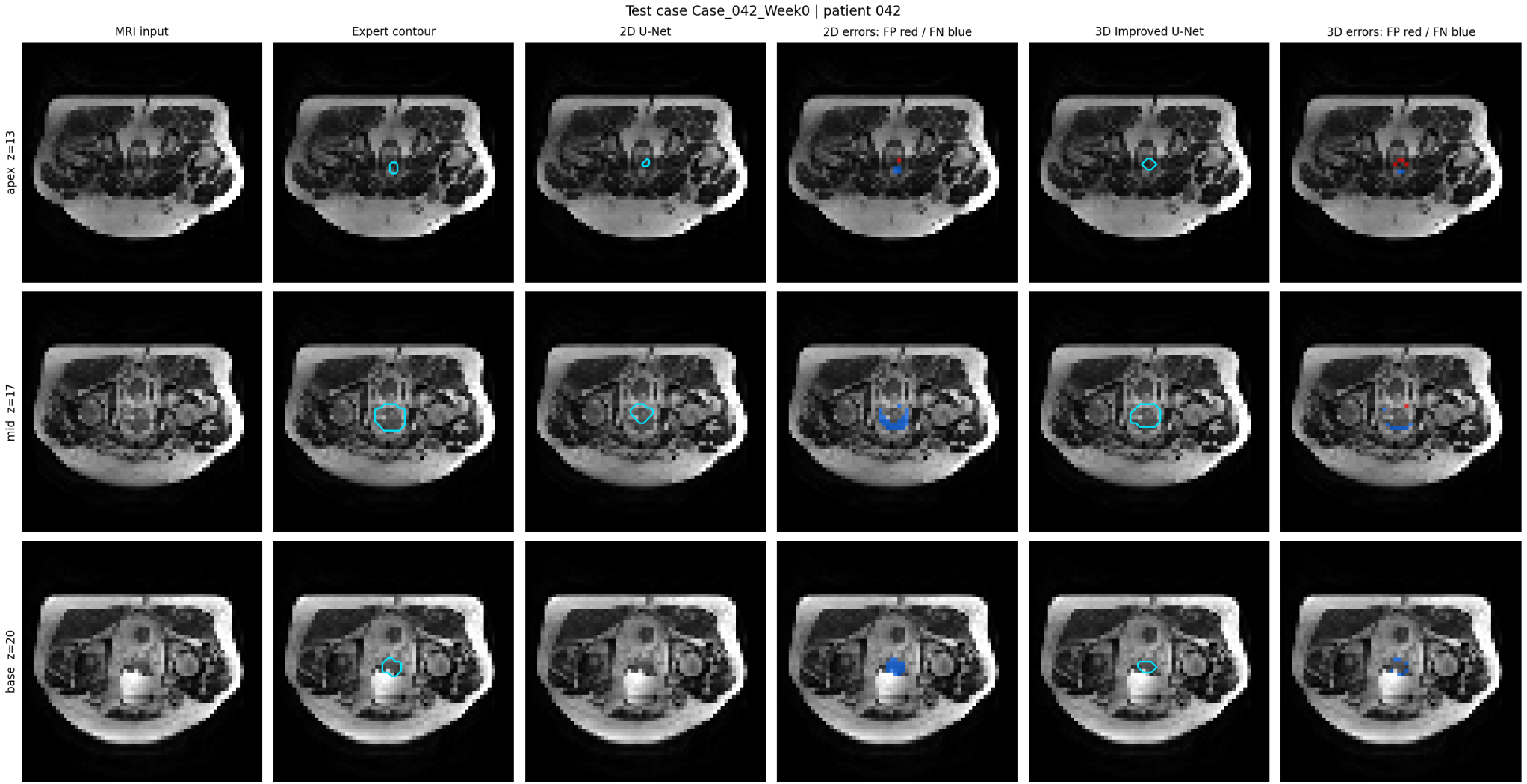


A counterexample: 3D underfills the upper mid-gland boundary. Base Dice falls from 0.517 to 0.399 and HD95 rises from 9.36 to 13.42 mm.

Panel source: repository figure. Red = false positive; blue = missed expert prostate. Image-grid panel; numeric Dice is native-grid.

Appendix A.4 | Case_042_Week0

Native Dice: 2D to 3D = 0.419 to 0.809 | full-field models only

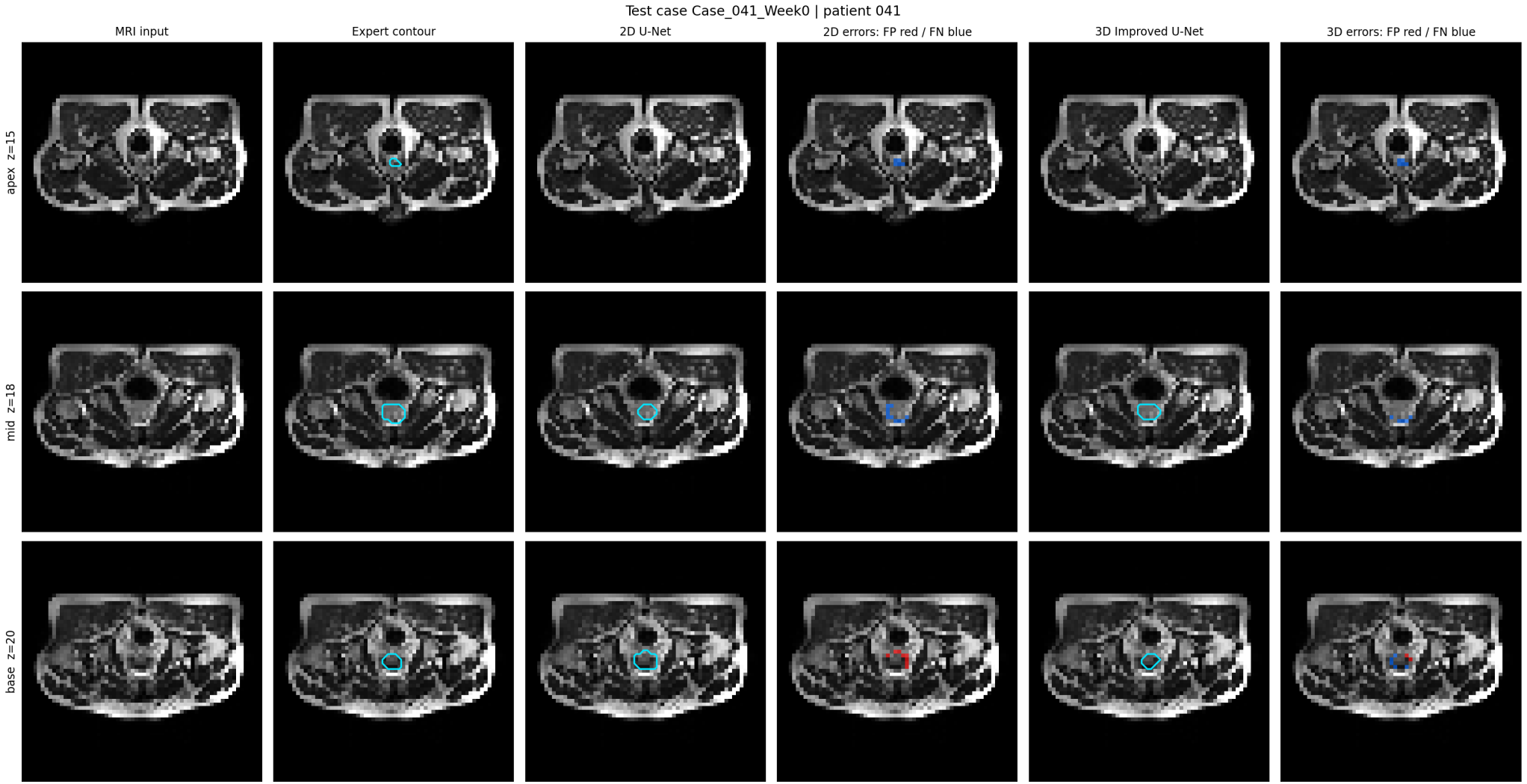


2D misses the base entirely (native base Dice 0); 3D recovers most of it (0.800), though missed pixels remain. Review blank base slices.

Panel source: repository figure. Red = false positive; blue = missed expert prostate. Image-grid panel; numeric Dice is native-grid.

Appendix A.5 | Case_041_Week0

Native Dice: 2D to 3D = 0.513 to 0.745 | full-field models only



Both models miss the tiny apex (native apex Dice 0). Stronger overall 3D overlap does not remove the edge-slice failure.

Panel source: repository figure. Red = false positive; blue = missed expert prostate. Image-grid panel; numeric Dice is native-grid.